Lab 8 — Mirror, Stretch, Scale and Arrays

WSQ Technical Drawing with AutoCAD (TGS-2021005538) · Topic 03: Modifying Objects

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Objective

LO2/LO3 — reproduce patterned and symmetric geometry (A3, K3)

Goal

Produce practice drawing EX-6 by drawing half of the symmetric design and mirroring it, then distributing the hole pattern with rectangular and polar arrays, and adjusting sizes with STRETCH and SCALE.

You'll produce

The completed EX-6 drawing with its symmetric outline and evenly arrayed hole pattern.

Commands / tools: MIRROR, STRETCH (crossing window), SCALE (Reference), ARRAYRECT, ARRAYPOLAR

Files in this lab

- ex6.dwg
- ex6-drawing-sheet.pdf
- ex6.png

Step-by-step

1. Open the EX-6 sheet (ex6-drawing-sheet.pdf) and find its axis of symmetry
2. Draw one half of the outline, then mirror it across the centreline

   ```
   MI → select → axis pt1 → pt2 → N
   ```

3. Array one hole into the rectangular pattern (rows, columns, spacing from the sheet)

   ```
   AR → R → rows/cols/spacing
   ```

4. Distribute the bolt holes around the centre with a polar array

   ```
   AR → PO → centre → items → 360
   ```

5. Stretch the slot to its dimensioned length with a crossing window, scale a feature with Reference

   ```
   S → crossing → base → distance  ·  SC → R
   ```

6. Verify against ex6.dwg and save

   ```
   SAVEAS ex6-mine.dwg
   ```

Check your work

The two halves are perfectly symmetric about the centreline, array counts and spacings match the sheet, and the stretched slot measures the dimensioned length.

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